KubeCon CloudNativeCon North America 2026

Volcano for the Agentic AI Era:

Unified Scheduling Across Training, Inference, and Agents

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Engineering Manager at Pinterest
Apache YuniKorn PMC member



Speakers

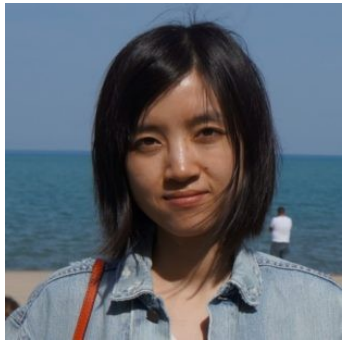


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CNCF TOC Alternate (GB)
Cloud native and AI infrastructure

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Data processing infrastructure
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01 Unified scheduling

Different paths. Shared infrastructure.

01 Unified scheduling

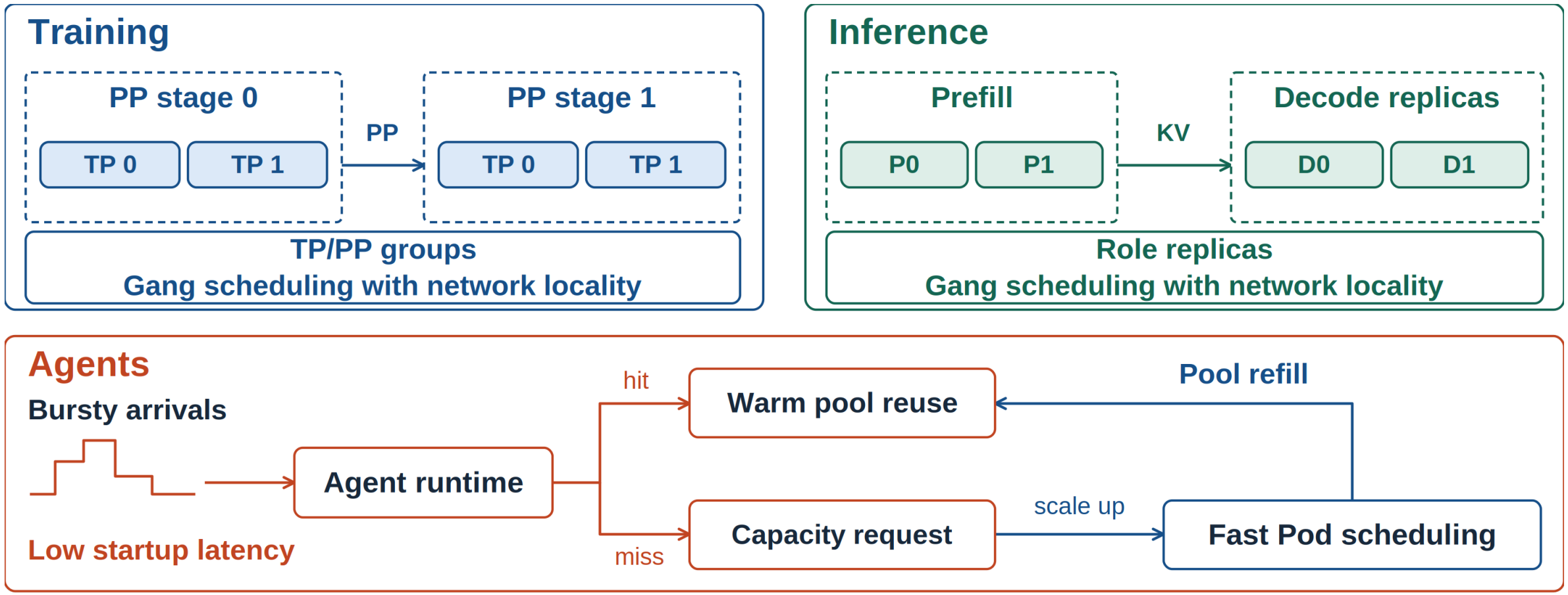
02 Multi-level gang & network topology

03 Ecosystem

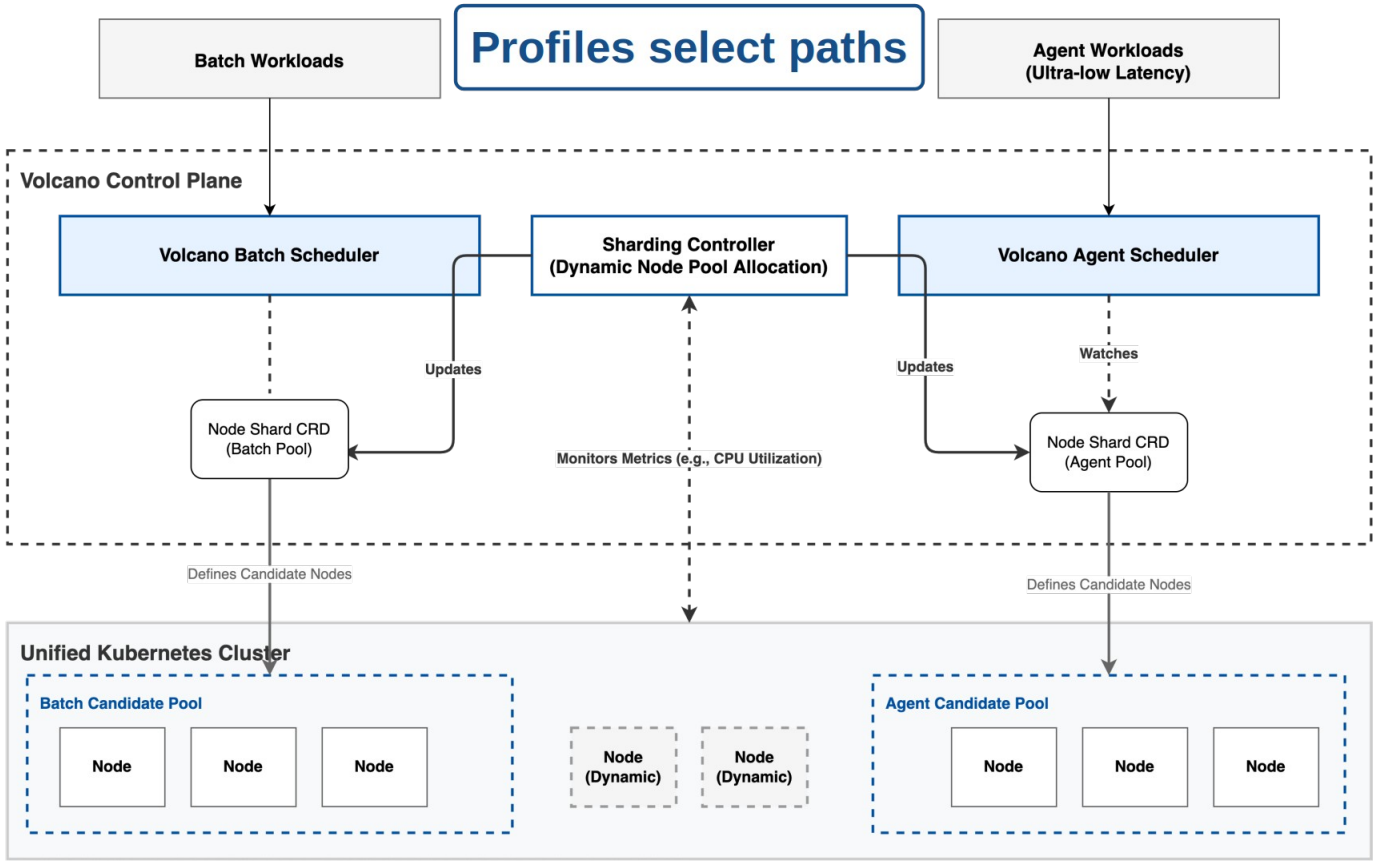
04 Benchmark framework & progress



Different Workloads, Shared Infrastructure



Unified Scheduling Framework



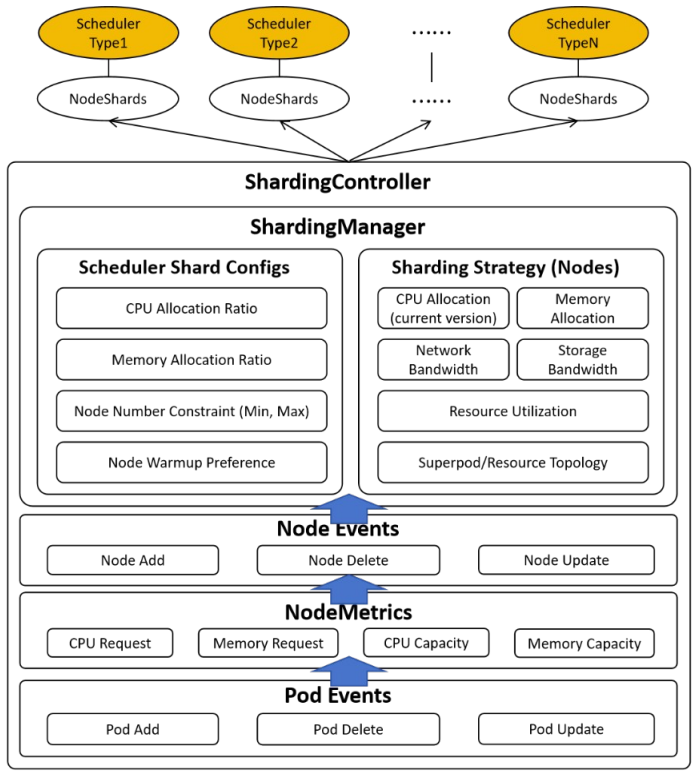
Sharding Controller

Live node state
Dynamic pool allocation

NodeShard

Candidate nodes
for each scheduling path

Dynamic Node Sharding



Node metrics + strategy
Compute desired node allocation

NodeShard spec and status

```
spec:
  nodesDesired:
  - node1
  - node2
status:
  nodesInUse:
  - node0
  - node1
  nodesToRemove:
  - node0
  nodesToAdd:
  - node2
```

Strategy result
Desired nodes

Scheduler state
Nodes in use

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02 Multi-level gang & network topology

From available nodes to runnable workloads.

01 Unified scheduling

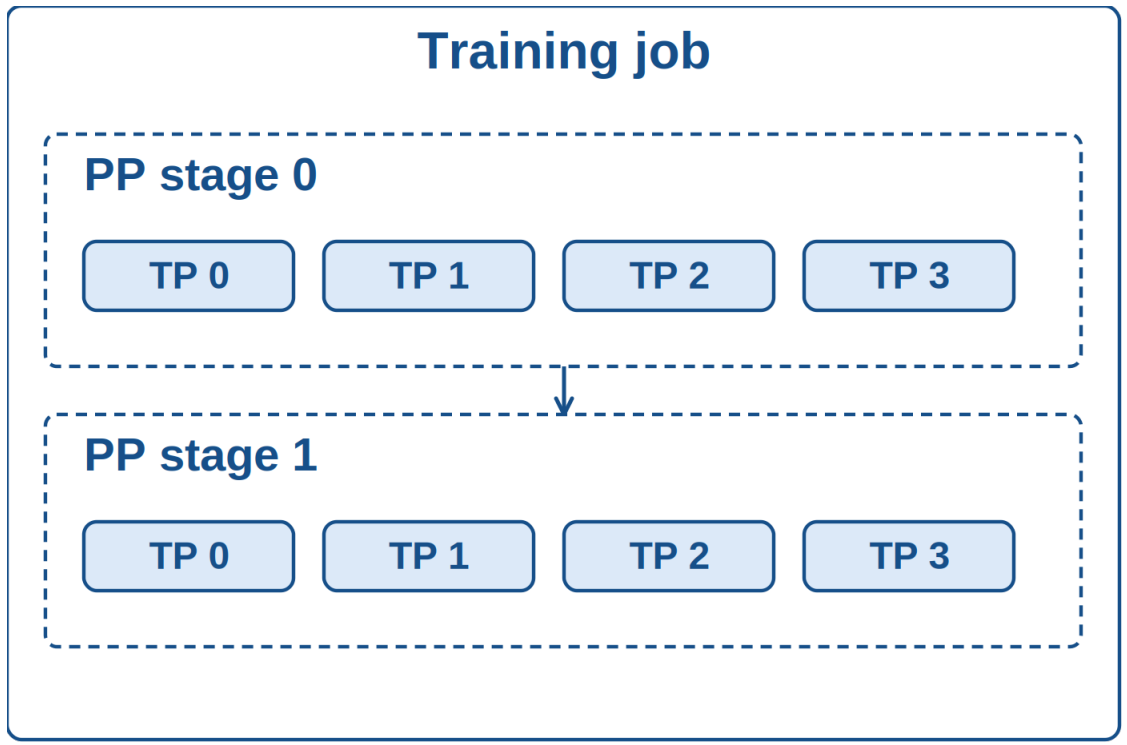
02 Multi-level gang & network topology

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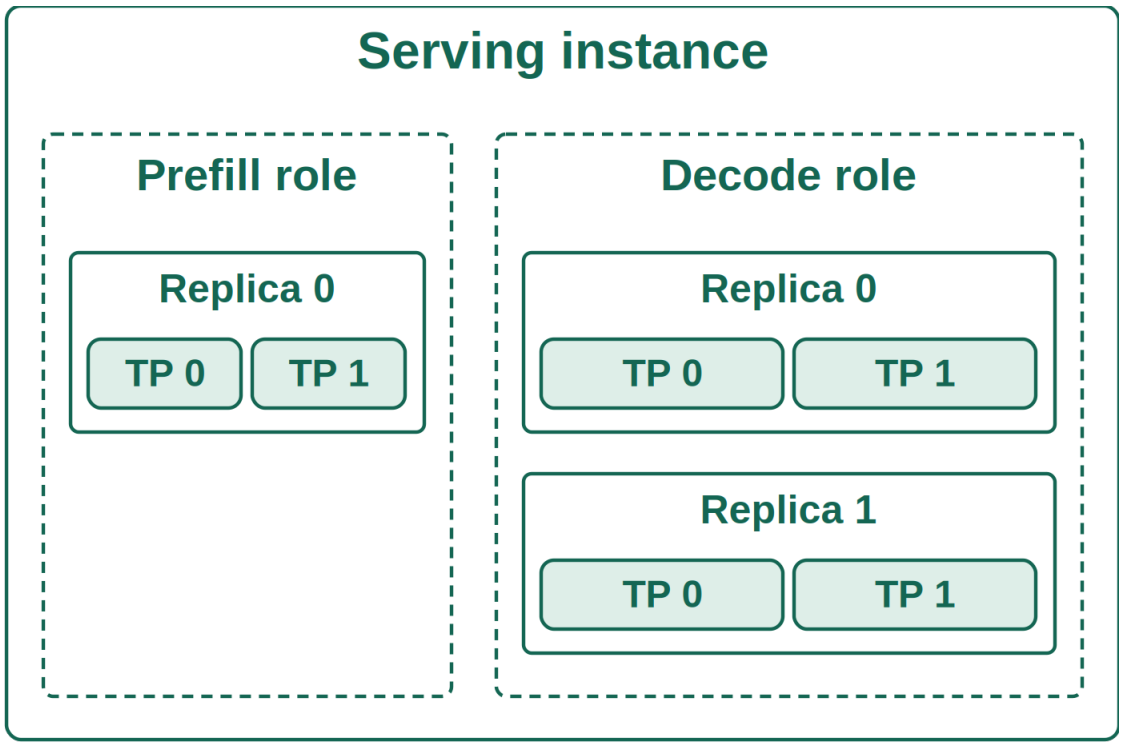
04 Benchmark framework & progress



Training and Inference Groups

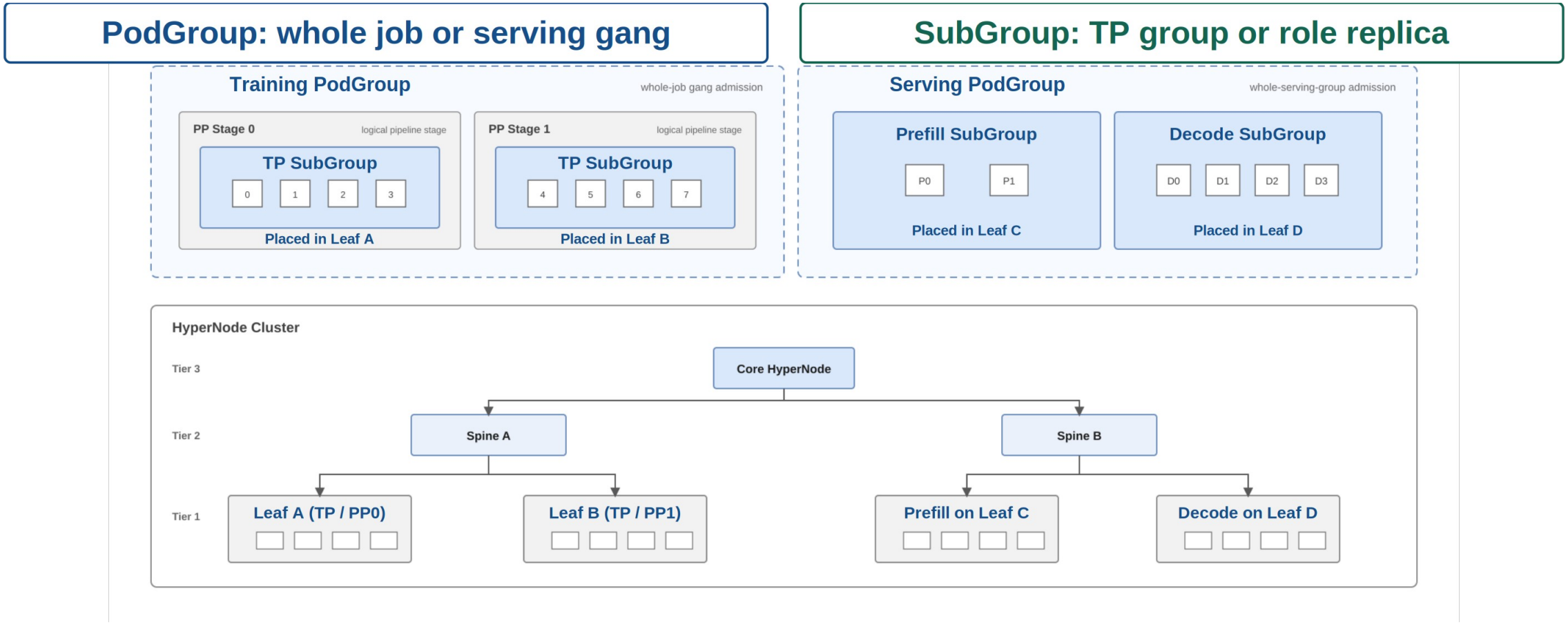


Complete TP groups within a job

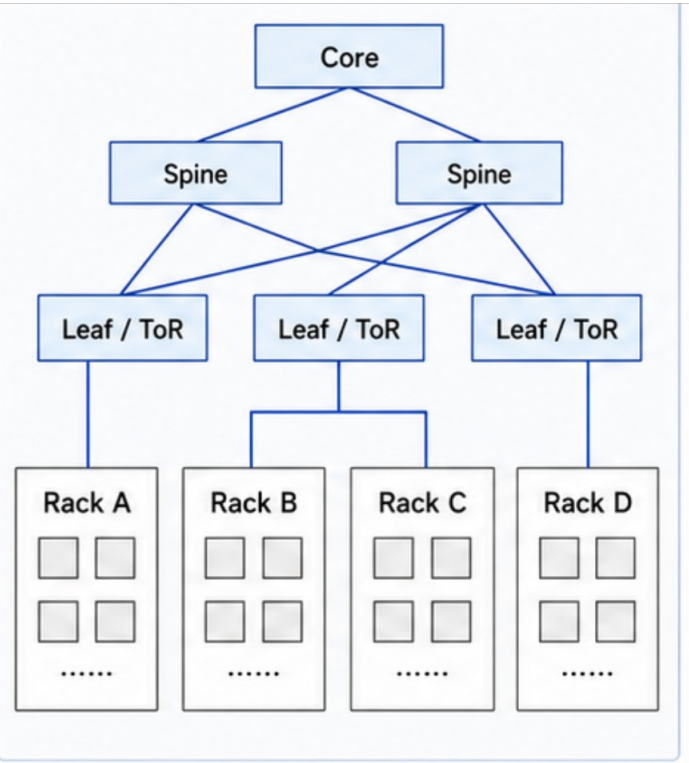


Complete replicas within a serving gang

Multi-Level Gang Scheduling



HyperNode: Network Topology



Parent HyperNode.spec

```
tier: 2
tierName: spine
members:
- type: HyperNode
  selector:
    exactMatch:
      name: rack-a
```

Groups child HyperNodes

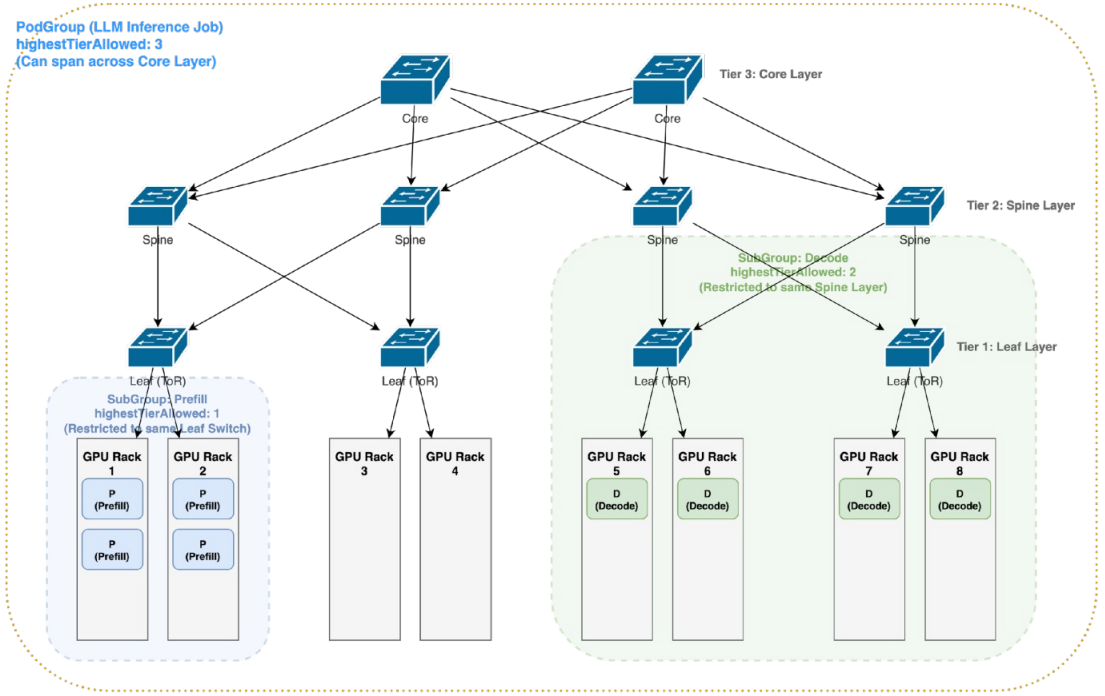
Leaf HyperNode.spec

```
tier: 1
tierName: leaf
members:
- type: Node
  selector:
    labelMatch:
      matchLabels:
        rack: rack-a
```

Groups Kubernetes nodes

Place Groups at Different Topology Levels

Different SubGroups, different topology constraints



Prefill at leaf tier

Decode at spine tier

PodGroup.spec (example)

```
minMember: 8
networkTopology:
  mode: hard
  highestTierName: core
subGroupPolicy:
- name: prefill
  subGroupSize: 4
  networkTopology:
    highestTierName: leaf
- name: decode
  subGroupSize: 4
  networkTopology:
    highestTierName: spine
```

Gang-Aware Preemption

Incoming gang S
`priority: 100`

S0 S1 S2 S3

`minMember: 4`
`highestTierName: spine`

Pod-level victims

HyperNode West
Each training gang needs both Pods

- Job A (priority 10): ~~A0~~ A1
- Job B (priority 20): ~~B0~~ B1
- Job C (priority 30): ~~C0~~ C1
- Job D (priority 40): ~~D0~~ D1

4 training jobs interrupted

Victim priority order

A (10): A0 A1
B (20): B0 B1
C (30): C0 C1
D (40): D0 D1

Simulate within HyperNode West

Try victim gangs A and B
S0 S1 S2 S3

No fit

No eviction

Fits

Commit

Feasible placement

HyperNode West

Job C: C0 C1
Job D: D0 D1

Gang S
S0 S1 S2 S3

A and B evicted as gangs

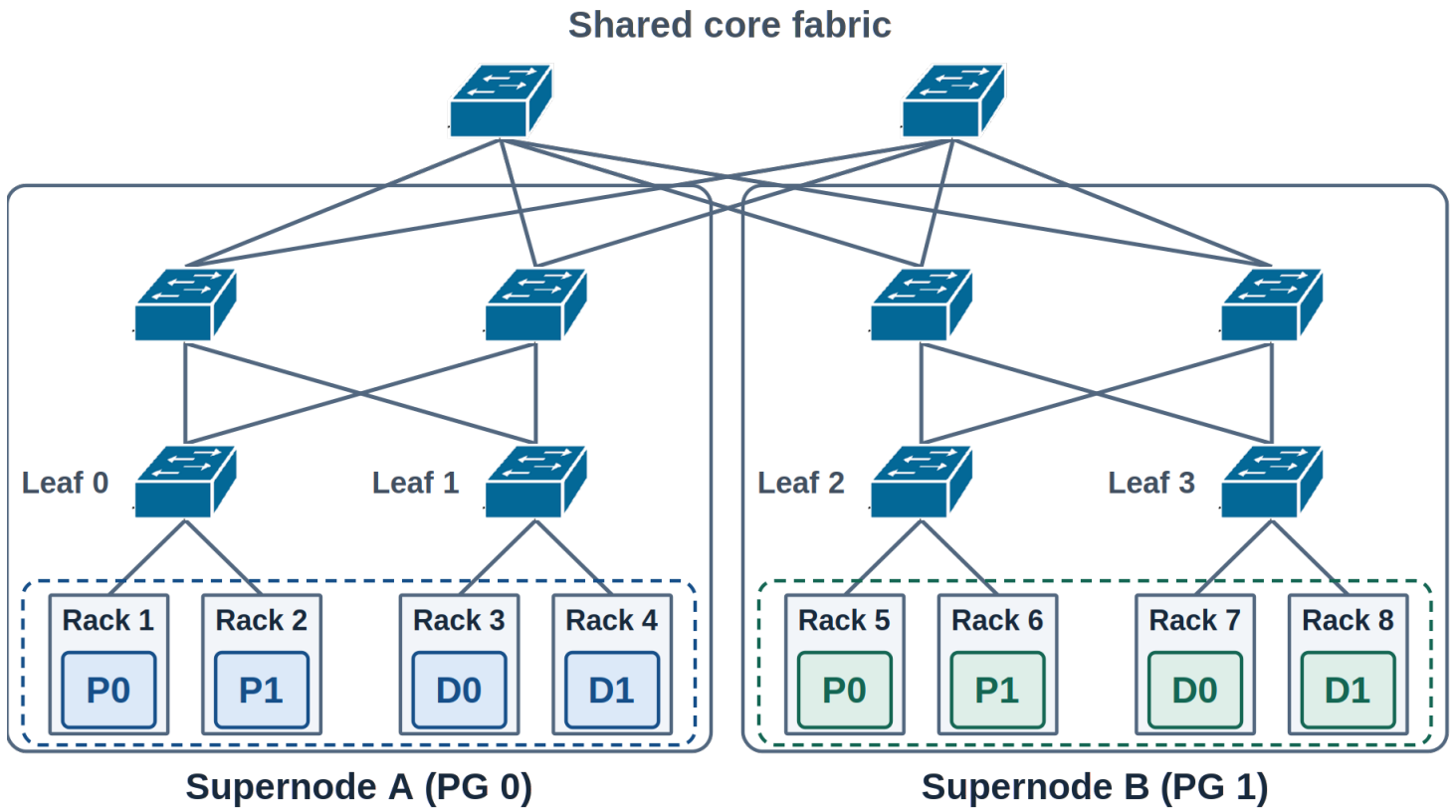
HyperNode East unchanged

Job E: E0 E1
Job F: F0 F1

PodGroup Anti-Affinity

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Separate matching PodGroups across supernodes



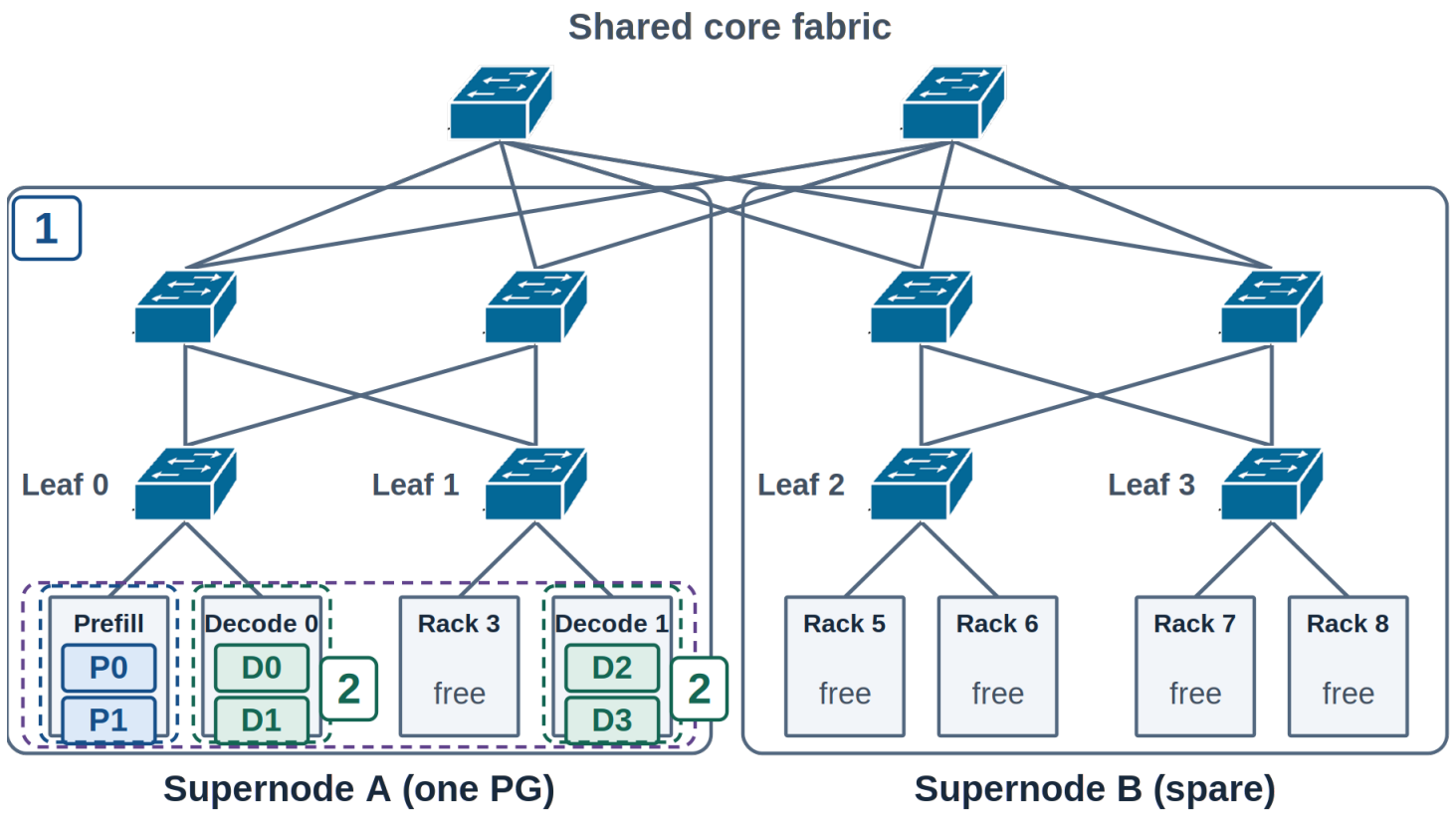
PodGroup.spec (key fields)

```
networkTopology:
  mode: hard
  highestTierName: supernode
topologyAffinity:
  podGroupAntiAffinity:
    required:
    - podGroupSelector:
        matchLabels:
          app: model-a
      topologyTierName: supernode
```

PG label: app=model-a

Separate supernodes with PodGroup anti-affinity

SubGroup Affinity and Anti-Affinity



Planned

1 Affinity within a supernode

Prefill + Decode

2 Anti-affinity across leaves

Decode replica 0 / replica 1

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03 Ecosystem

Workload controllers carry scheduling intent.

01 Unified scheduling

02 Multi-level gang & network topology

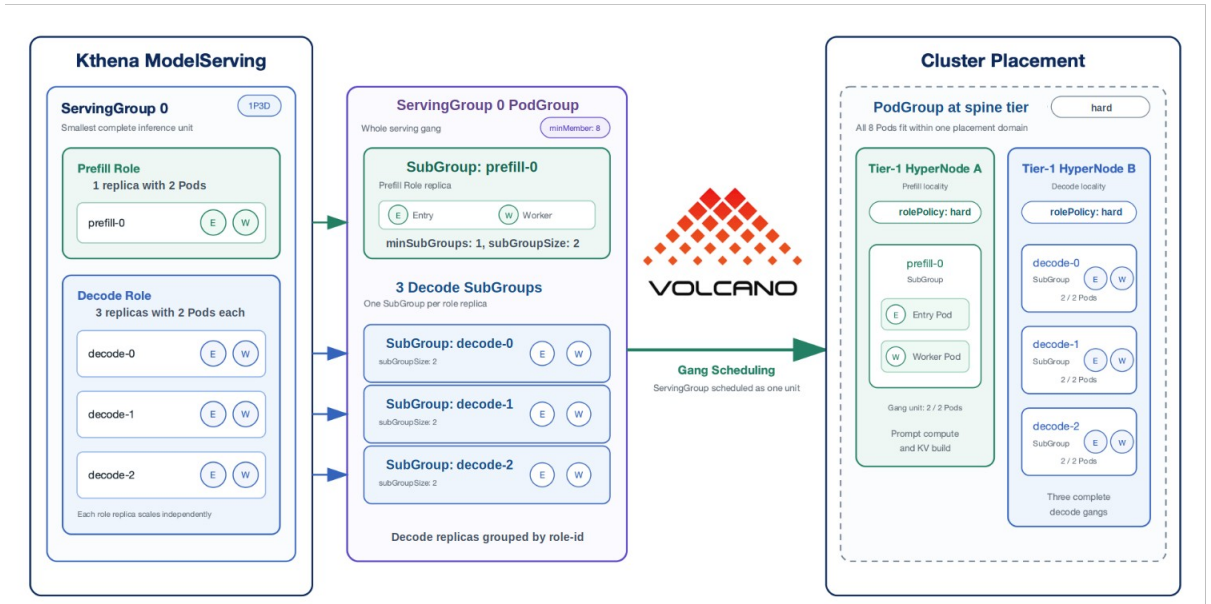
03 Ecosystem

04 Benchmark framework & progress



Kthena: Group and Role Topology

Kthena



Group: spine / Role replica: leaf

ModelServing.spec (excerpt)

```yaml
schedulerName: volcano
replicas: 1
template:
  gangPolicy:
    minRoleReplicas:
      prefill: 1
      decode: 3
  networkTopology:
    groupPolicy:
      mode: hard
      highestTierName: spine
    rolePolicy:
      mode: hard
      highestTierName: leaf
```

Kthena: Role Replica Mapping

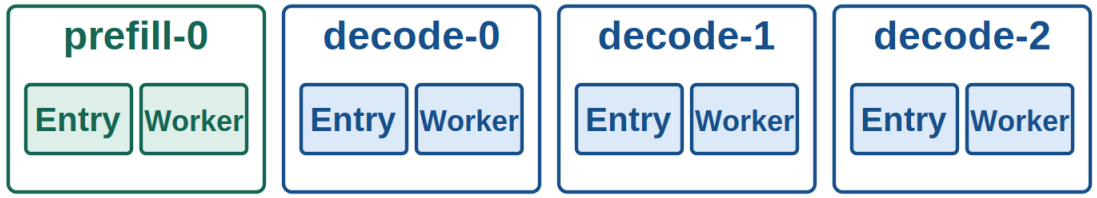
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Kthena

ModelServing.spec.template

```
roles:
- name: prefill
  replicas: 1
  workerReplicas: 1
- name: decode
  replicas: 3
  workerReplicas: 1
```

ServingGroup: 4 replicas, 8 Pods

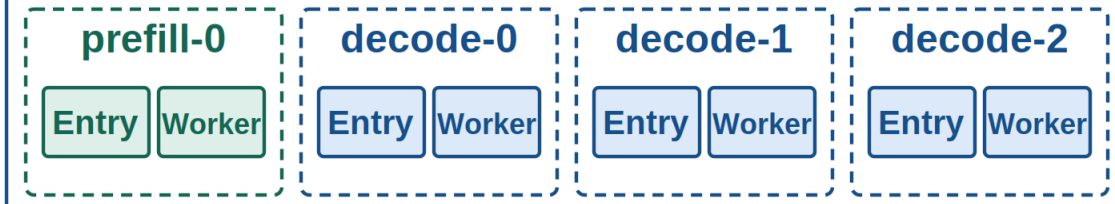


Volcano

PodGroup.spec (role policies)

```
minMember: 8
subGroupPolicy:
- name: prefill
  minSubGroups: 1
  subGroupSize: 2
- name: decode
  minSubGroups: 3
  subGroupSize: 2
```

PodGroup: 4 SubGroups, 8 Pods



KubeRay: Multi-Host Replica Mapping

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KubeRay

RayCluster.spec

```
enableInTreeAutoscaling: true
workerGroupSpecs:
- groupName: worker
  minReplicas: 2
  replicas: 2
  maxReplicas: 4
  numOfHosts: 2
```

RayCluster: 1 head + 4 worker Pods



Volcano

PodGroup.spec (head and workers)

```
minMember: 5
subGroupPolicy:
- name: headgroup
  minSubGroups: 1
  subGroupSize: 1
- name: worker
  minSubGroups: 2
  subGroupSize: 2
```

PodGroup: 3 SubGroups, 5 Pods



Grove: PodGang Mapping

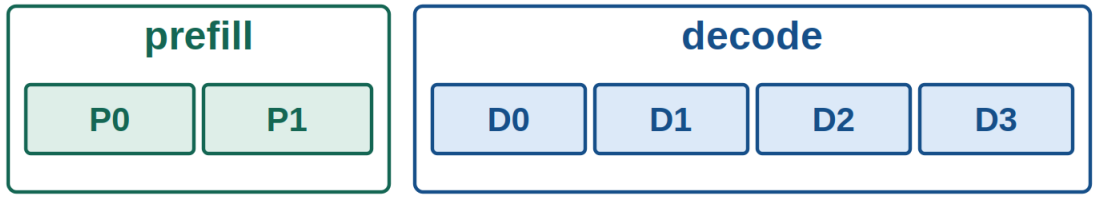
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Grove

PodGang.spec

```
podGroups:
- name: prefill
  minReplicas: 2
- name: decode
  minReplicas: 4
```

PodGang: 2 groups, 6 Pods



Volcano

PodGroup.spec (group policies)

```
minMember: 6
subGroupPolicy:
- name: prefill
  minSubGroups: 1
  subGroupSize: 2
- name: decode
  minSubGroups: 1
  subGroupSize: 4
```

PodGroup: 2 SubGroups, 6 Pods



Ecosystem Roadmap





RoleBasedGroup (RBG)

Integration in progress

PodGroup and SubGroup integration

HyperNode tree queries



Kubeflow

Planned

Training group structure

Network topology constraints



LeaderWorkerSet (LWS)

Planned

Multi-role and multi-group support

Relationships across groups

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04 Benchmark framework & progress

Measure behavior across workload shapes.

01 Unified scheduling

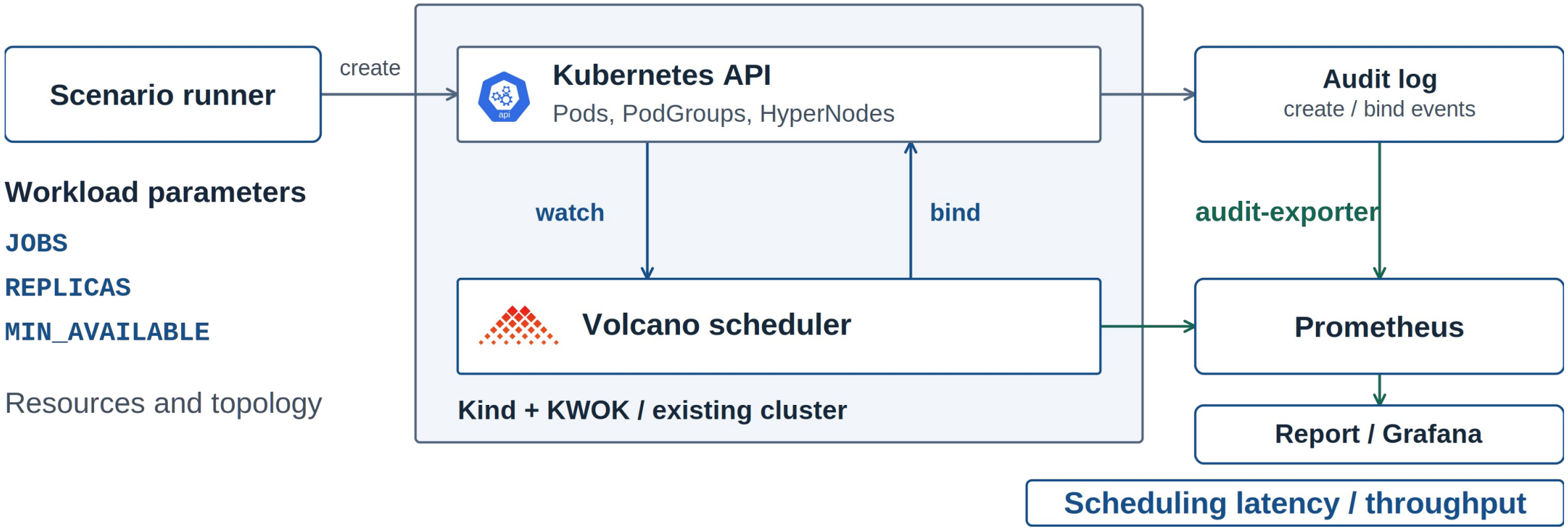
02 Multi-level gang & network topology

03 Ecosystem

04 Benchmark framework & progress



Scheduling Benchmark Framework

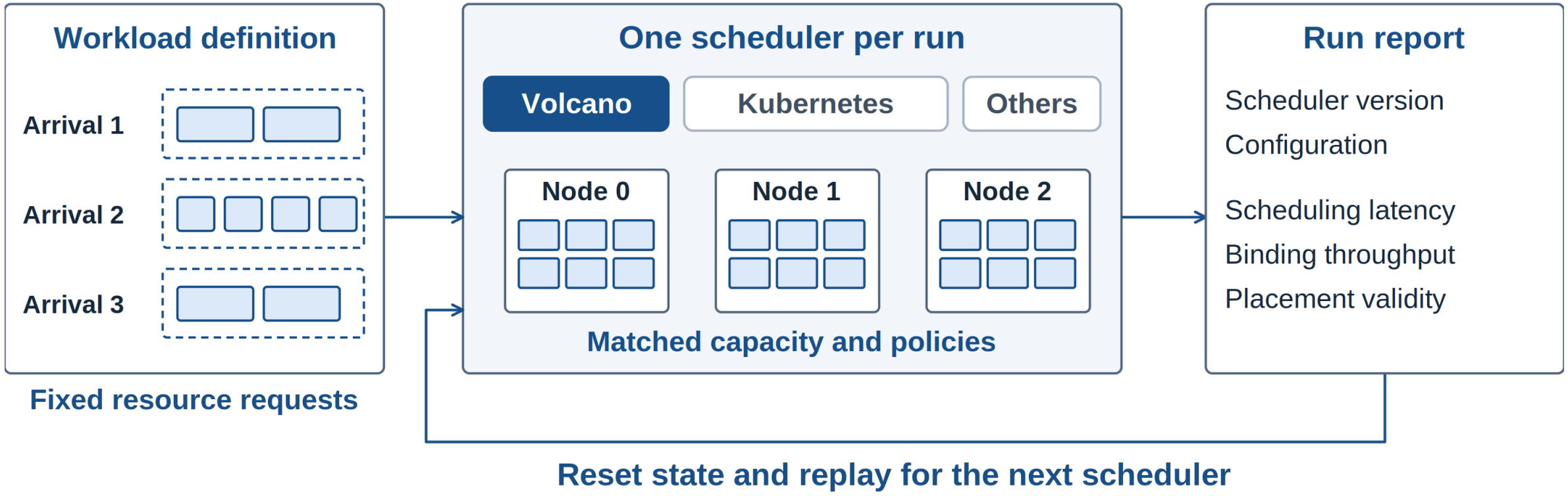


Cross-Scheduler Benchmarks

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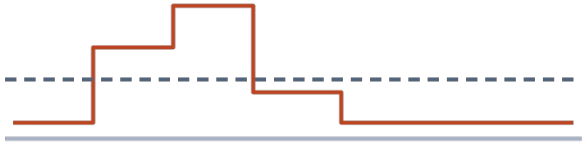
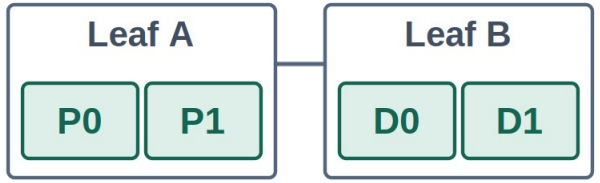
One workload, independent runs

Comparisons planned

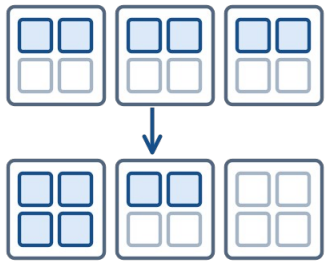


Workload and Topology Test Matrix

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Scenario		Variables	Metrics
Bursts		**Arrival rate and capacity** Pool capacity requests	**Tail latency** Backlog recovery
Gang size		**Group size and fragmentation** Large / small gangs	**Complete gangs** Binding throughput
Topology		**Tier limits and domains** Gang with anti-affinity	**Valid placement** Gang latency

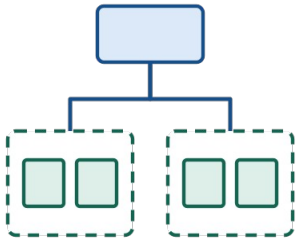
v1.17 Roadmap



Resource defragmentation

PodGroup-level relocation

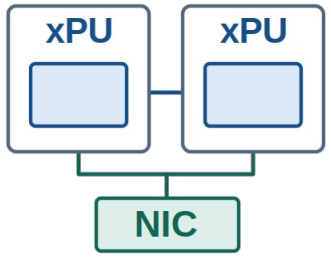
Consolidate GPU capacity



HyperNode and SubGroups

kube-scheduler extender

SubGroup affinity / anti-affinity



xPU Topology and DRA

Generic xPU topology

DRA binpack

DRA-backed extended resources

DRA preemption and reclaim

DRANET network allocation



Ecosystem integration

Kubeflow, KubeRay, LWS

Gang and topology support

Further work on preemption, queue fairness and performance

Volcano Users



Thank you

Questions & contributions

volcano.sh

GitHub

Slack